School of Computer and Systems Sciences
Jawaharlal Nehru University, New Delhi 110067, India
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The Editors
Intelligent Oncology

Dear Editors,

The authors submit the manuscript "Patient-level differential privacy, membership leakage and explanation fidelity in federated brain tumour MRI classification" for consideration as a full-length research article.

The manuscript addresses a question that affects any multi-hospital artificial intelligence model in neuro-oncology, namely what a formal privacy guarantee costs and what it actually protects. On a deduplicated three-source brain tumour MRI benchmark, a mirror-aware check found 66 contaminated test images that perceptual hashing had missed. In federated models, record-level differential privacy at $\varepsilon = 8$ left a likelihood-ratio membership attack with no rate significantly above chance once members and non-members were matched in source and class, for a balanced-accuracy cost of 0.076, against 0.130 in central training, whereas a per-patient guarantee needed $\varepsilon = 64$. Without privacy the same attack separated training from test patients with an AUC of 0.805, and an unclipped noise mechanism used in prior work still exposed patients. Grad-CAM localisation and cross-seed consistency declined with privacy in step with accuracy, and the accuracy lost under privacy fell mainly on meningioma and no-tumour sensitivity.

The work fits the journal's scope in artificial intelligence for oncology imaging and builds on themes the journal has published. Ahmed et al. reviewed artificial intelligence for brain tumour detection, grading and response evaluation (Intell Oncol 2026;2(3):100071, doi:10.1016/j.intonc.2026.100071), and Jung et al. applied deep learning to small brain metastases (Intell Oncol 2026;2(2):100049, doi:10.1016/j.intonc.2026.100049). Li et al. reviewed cryptographic privacy-preserving medical image processing (Intell Oncol 2026;2(4):100083, doi:10.1016/j.intonc.2026.100083), and Cai discussed transparency and privacy preservation in medical artificial intelligence (Intell Oncol 2025;1(2):172-175, doi:10.1016/j.intonc.2025.03.006). The manuscript complements these papers with a measured privacy, utility and explanation audit for federated neuro-oncology models.

The manuscript is original, has not been published, and is not under consideration elsewhere. All authors approved the submission. The authors declare no competing interests, and the work received no specific funding. Code, frozen results and figure data are available in a public repository [CHECK: repository visibility and archive DOI].

Two potential reviewers are suggested in the submission system [CHECK: names, institutional e-mail addresses, no conflicts].

Yours sincerely,

Poonguzhali N (corresponding author)
School of Computer and Systems Sciences, Jawaharlal Nehru University, New Delhi 110067, India
poonguzhali@jnu.ac.in, telephone [CHECK]

On behalf of Shashi Ranjan and Poonguzhali N